Calculation for power received according to path loss:

$$P_{rx}=P_{tx}+G_{rx}+G_{tx}+20\log\frac{4\pi df}{c}$$

this can be explained in the following paper page ٧:

https://www.slideshare.net/irjetjournal/statistical-analysis-of-received-power-in-an-antenna-down-tilt-on-cellular-networks?from_search=٣#٧

the following table is a SINR CQI mapping table suggessted:

A	B	C	D	E	F	G
%SNR CQI mapping table						
%Lower_limit_S	Upper_limit_SNI	CQI_index	Bits/sym	Modulation	Code rate	efficiency
% -20	-7	0			out of range	
% -6	-5	1	2	QPSK	78	0.1523
% -4	-3	2	2	QPSK	120	0.2344
% -2	-1	3	2	QPSK	193	0.377
%	0	4	2	QPSK	308	0.6016
% 1	3	5	2	QPSK	449	0.877
% 4	6	6	2	QPSK	602	1.1758
% 7	8	7	4	16QAM	378	1.4766
% 9	10	8	4	16QAM	490	1.9141
%	10	9	4	16QAM	616	2.4063
% 11	12	10	6	64QAM	466	2.7305
% 13	14	11	6	64QAM	567	3.3223
% 15	16	12	6	64QAM	666	3.9023
% 17	18	13	6	64QAM	772	4.5234
% 19	20	14	6	64QAM	873	5.1152
%	>20	15	6	64QAM	948	5.5547